Actividad 4 (Evaluación)

Ejercicio 1.

$x = [0 \text{ a } 5]$

$f(x) = 2\sin(x^2)$

$x(t) = t$ $\qquad$ $\dot{x}(t) = 1$

$y(t) = 2\sin(t^2)$ $\qquad$ $\dot{y}(t) = 4t\cos(t^2)$

condiciones iniciales

$x(0) = 0$

$y(0) = 2\sin(0^2) = 0$

$\theta(0) = \tan^{-1}(4(0)\cos(0^2)) = 0$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v(t) = \sqrt{1 + (4t\cos(t^2))^2}$$

$$\theta(t) = \tan^{-1}\left(\frac{4t\cos(t^2)}{1}\right)$$

$$w(t) = \frac{d(\tan^{-1}(\overbrace{4t\cos(t^2)}^{4(\cos(t^2) - 2t^2\sin(t^2))}))}{dt} = \frac{4(\cos(t^2) - 2t^2\sin(t^2))}{1 + (4t\cos(t^2))^2}$$

Trayectoria 2

$x = [-4 \text{ a } 4]$

$f(x) = x^2 + y^2 - 16$

condiciones iniciales

$x(0) = 0$

$y(0) =$

$x(t) = t$ $\qquad$ $\dot{x}(t) = 1$

$y(t) = x^2 + y^2 - 16$ $\qquad$ $\dot{y}(t) = 2x$

$$v(t) = \sqrt{1 + (2x)^2}$$

$$\theta(t) = \tan^{-1}(2x)$$

$$w(t) = \frac{d(\tan^{-1}(\overbrace{2x}^{2}))}{dt} = \frac{2}{1 + (2x)^2}$$

Trayectoria 3

$x = [-6 \text{ a } 6]$

$$f(x) = \begin{cases} 2x & x \leq -1 \\ 2x+1 & -1 < x < 1 \\ -x+4 & 1 \leq x < 4 \\ x-1 & x \geq 4 \end{cases}$$

tramo 1

$[-6 \quad -1]$

$2x$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = 2x \qquad \dot{y}(t) = 2$

condiciones iniciales

$x(0) = 0$

$y(0) = 0$

$\theta(0) = \tan^{-1}(2) = 1.107$

$$v(t) = \sqrt{1 + 2^2}$$

$$\theta(t) = \tan^{-1}(2)$$

$$\omega(t) = \frac{d\theta}{dt} = \frac{d(\overbrace{\tan^{-1}(2)}^{0})}{dt} = \frac{0}{1 + (2)^2} = 0$$

tramo 2

$(-1 \quad 1]$

$2x + 1$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = 2x + 1 \qquad \dot{y}(t) = 2$

$$v(t) = \sqrt{1 + 2^2}$$

$$\theta(t) = \tan^{-1}(2)$$

$$\omega(t) = \frac{d(\tan^{-1}(2))}{dt} = \frac{0}{1 + (2)^2}$$

tramo 3

$[1 \quad 4)$

$-x + 4$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = -x + 4 \qquad \dot{y}(t) = -1$

$$v(t) = \sqrt{1 + (-1)^2}$$

$$\theta(t) = \tan^{-1}(-1)$$

$$\omega(t) = 0$$

tramo 4

$[4 \quad 6]$

$x - 1$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = x - 1 \qquad \dot{y}(t) = 1$